AT6668 GPS Velocity Investigation on M5Stack GPS Unit

1. Executive Summary

This document summarizes a focused investigation into the GPS velocity outputs exposed by the tested M5Stack GPS unit on an AT6668-based receiver path. The purpose was to determine whether the module could provide a higher-quality or more explicit velocity channel than the existing NMEA Speed Over Ground (`SOG`) already used by the firmware.

The final production decision is narrow and specific to the tested module, tested firmware, and tested command path. `DHV` is not usable as the primary velocity source because it updates at approximately `1 Hz`, while the main GPS fix path updates at approximately `10 Hz`. Periodic `NAV-PV` is not available through the tested `CFG-PRT -> CFG-MSG` path, because the module accepts `CFG-PRT` but NACKs `CFG-MSG` for periodic `NAV-PV` enable. As a result, `gps_sog_kmh` remains the only viable production velocity source at `10 Hz`.

This report is intended both as an internal engineering record and as a vendor-facing escalation package for M5Stack. The key value of the work is not a new filter behavior, but a reproducible evidence chain: `.bin` logs, MQTT traces, and offline replay all converge on the same conclusion, and the retained `202 B` SD record format preserves that evidence for future module comparisons.

2. System Under Test

Module context

- Vendor product: M5Stack GPS unit using an AT6668-based receiver path
- Host MCU: M5Stack Atom S3 / ESP32-S3
- GPS interface: UART1 at `115200` baud

Firmware context

- `v1.4.2`: baseline production path using NMEA GPS speed
- `v1.5.0`: explicit `NAV-PV` attempt and diagnostic expansion
- `v1.5.1`: `DHV` primary-path attempt
- `v1.5.2`: production path simplified back to `gps_sog_kmh`, while diagnostic fields were retained

Test environment

- Indoor static tests
- Outdoor static tests
- Standalone power-bank operation, without serial tether
- MQTT-based remote observation
- SITL / offline replay from SD logs

3. Objective

The investigation had three goals:

1. Determine whether `DHV` could replace NMEA `SOG` as a higher-quality or more explicit production velocity source.
2. Determine whether periodic binary `NAV-PV` could be enabled and used as a `10 Hz` production velocity channel.
3. Decide whether the existing `SOG` path should remain the production baseline.

For this project, a velocity source is considered **usable as a primary source** only if it:

- supports the effective `10 Hz` GPS update path
- is compatible with stationary detection
- is compatible with straight-line ZARU speed gates
- is compatible with ESKF speed correction
- is compatible with COG baseline logic

4. Instrumentation Added

SD / binary logging

The SD record was extended from `164 B` (`v1.4.2`) to `202 B` (`v1.5.x`) to preserve GPS source and timing diagnostics.

Additional fields retained in the `202 B` record:

Field	Size
`nav_speed2d`	4 B
`nav_s_acc`	4 B
`nav_vel_n`	4 B
`nav_vel_e`	4 B
`nav_vel_valid`	1 B
`gps_speed_source`	1 B
`nav_fix_us`	8 B
`dhv_gdspd`	4 B
`dhv_fix_us`	8 B
Total	**38 B**

What the extra 38 B buy

The extra `38 B` do not add new filter dynamics. They buy:

- deterministic offline proof of source availability
- deterministic offline proof of source selection
- repeatable SITL replay of GPS timing and source behavior
- vendor-facing evidence without requiring serial capture

At `50 Hz`, the extra `38 B` add:

- `1900 B/s`
- `6.84 MB/hour` (decimal MB)

This storage cost was judged small compared with the diagnostic value already extracted from the field data.

MQTT diagnostics

The following MQTT fields were added or used during the investigation:

- GPS timing: `gps_fix_us`, `gps_epoch`, `dhv_fix_us`, `nav_fix_us`
- Source visibility: `spd_src`, `nav_spd2d`, `dhv_spd2d`
- CASIC parser diagnostics: `dbg_sync`, `dbg_frames`, `dbg_ck_fail`, `dbg_other`, `dbg_navpv`, `dbg_navpv_vel`, `dbg_ovf`
- Last-frame diagnostics: `dbg_cls`, `dbg_id`, `dbg_len`, `dbg_ack_fid`, `dbg_ack_cls`, `dbg_ack_id`
- Configuration outcome diagnostics: `dbg_prt_ack`, `dbg_prt_nack`, `dbg_cfg_try`, `dbg_cfg_ack`, `dbg_cfg_nack`

`CFG-PRT` outcome is already visible in MQTT in the current firmware and therefore no longer requires a serial cable for outdoor verification.

SITL / offline tooling

The following tools were used or updated during the investigation:

- `Tool/bin_to_csv.py`
- `Tool/motec_exporter.py`
- `Tool/sitl_hal/sitl_replay.py`
- `Tool/sitl_hal/static_gate_debug.py`
- `Tool/sitl_hal/gps_speed_compare.py`
- `Tool/sitl_hal/static_bench_validator.py`
- `Tool/sitl_hal/static_bench_compare.py`

These tools allow offline reconstruction of:

- GPS timing and staleness behavior
- source selection behavior
- static-gate outcomes
- speed-channel comparisons

5. Methodology

`.bin` log analysis

Static and quasi-static `.bin` logs were replayed offline to evaluate:

- whether GPS speed alone could trigger stationary-gate failures
- whether outdoor RF quality changed the failure mode
- whether DHV or NAV-PV became active in the logs

MQTT trace analysis

MQTT traces were used because the outdoor workflow ran from a power bank, without a serial tether. They were used to measure:

- effective update rate of `gps_fix_us`
- effective update rate of `dhv_fix_us`
- whether `CFG-PRT` was ACKed
- whether `CFG-MSG` was ACKed or NACKed
- whether any periodic `NAV-PV` frames appeared after configuration

CASIC ACK/NACK interpretation

The investigation used the observed CASIC ACK/NACK fields:

- `dbg_cls`
- `dbg_id`
- `dbg_ack_fid`
- `dbg_ack_cls`
- `dbg_ack_id`

to determine whether the module:

- accepted `CFG-PRT`
- rejected periodic `NAV-PV` enable via `CFG-MSG`

Freshness / source validation

The work explicitly separated:

- parser correctness
- source frequency
- source selection policy

This prevented a false conclusion that `DHV` was unusable because of parser bugs or freshness policy. The evidence showed instead that `DHV` itself updated at about `1 Hz`.

Indoor vs outdoor separation

Indoor and outdoor logs were both used so that:

- RF degradation and multipath could be identified when present
- protocol limitations could be distinguished from bad reception

Outdoor static logs were especially important because they removed RF quality as the primary blocker while `DHV` and `NAV-PV` still failed to become viable production sources.

6. Evidence

6.1 Artifact inventory

Primary artifacts used in this report:

ID	Artifact	Role
A1	tel_11_speed_compare.txt	Speed-channel comparison on static bench data
A2	tel_20_static.txt	Outdoor static validation
A3	tel_21_static.txt	Outdoor static validation
A4	tel_20_gate.txt	Outdoor static gate-failure attribution
A5	tel_21_gate.txt	Outdoor static gate-failure attribution
M0	Telemetria.json	Initial MQTT symptom capture
M1	Telemetria1.json	MQTT timing capture after gps_fix_us instrumentation
M2	Telemetria2.json	MQTT timing capture after explicit PCAS03 test
M3	Telemetria3.json	MQTT protocol diagnostics after CFG-PRT -> CFG-MSG test

6.2 tel_11 speed-channel comparison

Source artifact: A1

Clean-dt (90–110 ms) comparison:

Metric	Module speed	Lat/lon/dt speed	kf_vel
Mean	1.861 km/h	3.434 km/h	1.647 km/h
Correlation with kf_vel	0.741	0.271	—
Ratio >= 2.0 km/h	35.8%	63.4%	27.3%

Observed conclusion:

- module-reported speed is substantially closer to kf_vel than position-derived speed
- lat/lon/dt-derived speed is much noisier than the module-reported speed
- therefore “compute speed from position” is not a valid improvement path on this hardware/log set

Classification:

- **Proven observation** based on log analysis

6.3 Outdoor static gating (tel_20 / tel_21)

Source artifacts: A2, A3, A4, A5

Key results:

Log	Static ratio	due_to_gps	Result
tel_20	100.0%	0	PASS
tel_21	97.5%	0	PASS

Observed conclusion:

- outdoor RF quality is not the blocker
- GPS-induced stationary-gate failures disappear outdoors
- the remaining blocker is protocol/channel behavior, not outdoor signal quality

Classification:

- **Proven observation** based on outdoor logs

6.4 MQTT timing evidence

Source artifacts: `M1`, `M2`

Extracted timing:

File	Mean `gps_fix_us` delta	Mean `dhv_fix_us` delta	Interpretation
`Telemetria1.json`	`~107339 us`	`~1000764 us`	GPS fix path ~10 Hz, DHV path ~1 Hz
`Telemetria2.json`	`~107588 us`	`~1000099 us`	GPS fix path ~10 Hz, DHV path ~1 Hz

Observed conclusion:

- `PCAS02` successfully drives the main GPS fix path at about `10 Hz`
- `DHV` does not follow that rate and remains at about `1 Hz`

Classification:

- **Proven observation** based on MQTT timing deltas

6.5 MQTT protocol diagnostics

Source artifact: `M3`

Relevant fields:

Field	Value
`dbg_prt_ack`	`1`
`dbg_prt_nack`	`0`
`dbg_cfg_try`	`1`
`dbg_cfg_ack`	`0`
`dbg_cfg_nack`	`1`
`dbg_navpv`	`0`
`dbg_navpv_vel`	`0`

Field	Value
`dbg_cls`	`5`
`dbg_id`	`0`
`dbg_ack_fid`	`0`
`dbg_ack_cls`	`6`
`dbg_ack_id`	`1`

Observed conclusion:

- `CFG-PRT` was accepted
- `CFG-MSG` for periodic `NAV-PV` was NACKed
- no periodic `NAV-PV` stream was observed afterward

Classification:

- **Proven observation** based on MQTT diagnostics

6.6 Initial live symptom (`Telemetria.json`)

Source artifact: `M0`

Observed behavior:

- `spd_src` oscillated between `0` and `2`
- `dhv_fix_us` changed at approximately `1 Hz`
- this was the first live symptom that `DHV` could not remain the active production source under the existing freshness policy

This trace predates the later `gps_fix_us` instrumentation and therefore is useful mainly as an early symptom record, not as the strongest timing proof.

Classification:

- **Proven observation**, but weaker than `Telemetria1/2/3`

7. DHV Findings

The following statements are supported by the evidence above:

- `DHV` is parsed correctly.
- `DHV` updates regularly.
- `DHV` updates at approximately `1 Hz`, not `10 Hz`.
- Explicit `PCAS03` formatting did not change the observed `DHV` rate.

This is sufficient to reject `DHV` as the primary `10 Hz` control source on the tested module.

Classification:

- **Proven observations**: parser works, `DHV` is present, `DHV` is ~`1 Hz`

- **Engineering conclusion**: `DHV` is not fit for use as the production velocity source in this firmware pipeline

8. NAV-PV Findings

The following statements are supported by the evidence above:

- passive CASIC parsing and ACK/NACK interpretation work
- binary output can be enabled on the port via `CFG-PRT`
- periodic `NAV-PV` enable via `CFG-MSG (0x06, 0x01)` is NACKed
- no periodic `NAV-PV` frames were observed afterward

This is sufficient to reject periodic `NAV-PV` as an available production channel on this module through the tested command path.

Classification:

- **Proven observations**: `CFG-PRT` ACK, `CFG-MSG` NACK, no `NAV-PV`
- **Engineering conclusion**: periodic `NAV-PV` is not available through the tested configuration path on the tested module

9. Production Decision

The production decision is:

- keep `gps_sog_kmh` as the sole production velocity source
- keep `DHV` and `NAV-PV` fields only for diagnostics
- keep the `202 B` record because the additional diagnostics have already proven their value

This decision is pragmatic:

- it preserves the working `10 Hz` production path
- it avoids further development time on channels that did not become viable
- it preserves the evidence needed for vendor escalation and future module comparisons

10. Vendor Questions / Requested Clarifications

The following clarifications are requested from M5Stack:

1. Is `DHV` intentionally rate-limited to approximately `1 Hz` on this module firmware, even when the main positioning path is configured for approximately `10 Hz`?
2. Is periodic `NAV-PV` unsupported on this product, disabled in this firmware build, or dependent on a different command sequence than the tested `CFG-PRT -> CFG-MSG` path?
3. Does this module expose any other supported high-rate velocity channel suitable for production use at approximately `10 Hz`?
4. If periodic `NAV-PV` is supported, what is the correct enable sequence and expected ACK behavior for this module specifically?

Appendix A — Proven observations vs engineering inference

Proven observations

- Module-reported speed tracks `kf_vel` better than position-derived speed in `tel_11`
- Outdoor static logs do not show GPS-driven stationary-gate failures
- Main GPS fix path updates at approximately `10 Hz`
- `DHV` updates at approximately `1 Hz`
- `CFG-PRT` is ACKed
- `CFG-MSG` for periodic `NAV-PV` is NACKed
- No periodic `NAV-PV` stream appears after the tested enable attempt

Engineering inference

- The tested AT6668 module/firmware likely imposes a practical `DHV` rate limitation on this path
- The tested AT6668 module/firmware likely does not support periodic `NAV-PV` through the tested command sequence

Appendix B — Record-size rationale

`v1.4.2` record size: `164 B`
`v1.5.2` record size: `202 B`
Increase: `38 B`

Those `38 B` are retained because they provide:

- source visibility
- source-timing visibility
- deterministic replay support
- vendor-facing, portable evidence

Without them, the firmware would still run, but this investigation would have been materially weaker and much harder to defend to a third party.

Appendix C — Reference artifacts

ID	Filename	Description
A1	`tel_11_speed_compare.txt`	Static speed-channel comparison report
A2	`tel_20_static.txt`	Outdoor static validation report
A3	`tel_21_static.txt`	Outdoor static validation report
A4	`tel_20_gate.txt`	Outdoor static gate debug report
A5	`tel_21_gate.txt`	Outdoor static gate debug report
M0	`Telemetria.json`	Initial MQTT capture showing DHV-selected symptom bursts
M1	`Telemetria1.json`	MQTT capture used to estimate GPS fix vs DHV update rate
M2	`Telemetria2.json`	MQTT capture after explicit `PCAS03` test

ID	Filename	Description
M3	Telemetria3.json	MQTT capture proving CFG-PRT ACK and CFG-MSG NACK
R1	Multimode_satellite_navigation_receiver_cn.pdf	Receiver protocol reference consulted during the investigation
R2	AT6668.pdf	Chip/module reference consulted during the investigation
R3	MAX2659.pdf	Front-end reference consulted during the investigation

Raw artifacts are retained by the author and can be shared with M5Stack on request.